



SCHOOL OF ADVANCED TECHNOLOGY

SAT301 FINAL YEAR PROJECT

Neuro-Physical Piano Synthesis: A Full-Pipeline System for Sustain Pedal Detection and Acoustic Rendering

Final Thesis

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Student Name : Xiaofei Chen

Student ID : 2252798

Supervisor : Dr. Shengchen Li

Assessor : Dr. Dechang Xu

Abstract

The sustain pedal is a fundamental component of piano performance, enabling complex sympathetic resonance and expressive tonal blending. However, accurately capturing and recreating these effects in digital environments remains a significant challenge. This project develops a comprehensive, full-pipeline system, termed Neuro-Physical Piano Synthesis (NPPS), designed to bridge the gap between abstract acoustic signals and physically coherent piano rendering.

The proposed system is divided into two primary stages: high-resolution pedal detection and physically-constrained acoustic synthesis. In the first stage, a deep neural network based on the U-Net architecture is implemented as a perception frontend. This module extracts frame-level pedal engagement probabilities from raw "dry" piano recordings. To ensure data quality, a rigorous data engineering pipeline was established using the MAESTRO v3.0.0 dataset, involving multimodal alignment and denoising of MIDI-ground-truth labels.

In the second stage, to address the "physical incoherence" common in black-box generative models—specifically high-frequency mechanical jitter and unnatural acoustic decay—a dual-level physical constraint interface is introduced. A mechanical layer utilizing a Local Context-Aware Schmitt Trigger (LCAST) and asymmetric inertia enforces kinematic boundaries on the neural predictions. Subsequently, an acoustic rendering engine employing a dual-stage energy dissipation model and an anti-aliasing harmonic exciter reconstructs the spectral bloom of the piano soundboard.

Experimental results demonstrate that the system achieves high precision in pedal tracking and reaches a state-of-the-art Fréchet Audio Distance (FAD) score of 9.88, significantly outperforming industrial baselines. The findings confirm that integrating mechano-acoustic causal chains into neural architectures is essential for achieving artifact-free and authentic musical synthesis.

Keywords : Piano Synthesis; Sustain Pedal Detection; Physical Constraints; Deep Learning; Differentiable Digital Signal Processing; Mechano-Acoustics

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Contents

Abstract	ii
Acknowledgements	iii
List of Tables	vii
List of Figures	viii
List of Acronyms	ix
1 Introduction	1
1.1 Motivation	1
1.2 Aims and Objectives	2
1.3 Literature Review	2
1.3.1 Piano Mechanics and Acoustic Resonance	2
1.3.2 Automatic Piano Transcription and Pedal Detection	3
1.3.3 Neural Audio Synthesis and DDSP	3
2 System Design and Methodology	4
2.1 System Overview	4
2.2 Data Engineering and Multimodal Alignment	4
2.2.1 Characteristics of the MAESTRO v3.0.0 Corpus	5
2.2.2 Surgical Extraction of Anechoic Dry Audio	5
2.2.3 Synthesis of the Privileged Wet Baseline	6
2.2.4 Time-Series Discretization and Algorithmic Alignment	6
2.2.5 Dynamic Segmentation and Memory Optimization	7
2.3 Neural Perception Module (PedalNet)	8
2.3.1 Acoustic Feature Extraction and Physical Resolution	8
2.3.2 Encoder: Downsampling and Abstract Representation	9
2.3.3 Decoder: Upsampling and Skip Connections	10

2.3.4	Temporal Projection and Probability Activation	10
2.4	Mechanical Constraint Interface: Tensorized Implementation	11
2.4.1	LCAST Tensorization via 1D Max Pooling	11
2.4.2	Asymmetric Inertia and ODE Discretization	12
2.4.3	Mechanism Noise Extrapolation	13
2.5	Acoustic Constraint Engine: High-Efficiency Rendering	13
2.5.1	Dual-Stage Energy Dissipation Modeling	13
2.5.2	Anti-Aliasing Harmonic Exciter	13
2.5.3	Complexity Reduction via FFT Convolution	14
3	Experimental Setup and Results	16
3.1	Experimental Configuration	16
3.1.1	Data Allocation and Environment	16
3.2	Training Dynamics and Hyperparameter Tuning	16
3.2.1	Addressing Class Imbalance via Weighted BCE	17
3.2.2	Optimization Strategy and Mixed Precision	17
3.2.3	Convergence Analysis	18
3.3	Evaluation Metrics: An Orthogonal Framework	18
3.3.1	Macroscopic Fidelity Metrics	18
3.3.2	Mechanistic Regularization Metrics	19
3.4	Objective Evaluation and Ablation Study	19
3.4.1	Analysis of the Privileged Industrial Baseline	19
3.4.2	Mechanical Overfitting Defense and Kinematic Legality	20
3.4.3	Acoustic Boundary and Spectral Bloom Decoupling	21
3.5	Mechanistic and Qualitative Analysis	21
3.5.1	Kinematic Trajectory Dynamics	21
3.5.2	Spectral Bloom Reconstruction and Anti-Aliasing	22
3.6	Subjective Perceptual Evaluation (MOS)	24
3.6.1	Web-Based Evaluation Infrastructure	24
3.6.2	Double-Blind Randomization Engineering	25
3.6.3	Listener Profile and Statistical Variance Analysis	25

4	Conclusion and Future Work	27
4.1	Conclusion	27
4.2	Future Work	28
4.2.1	Polyphonic Tracking and Complex Pedal Techniques	28
4.2.2	Real-time Processing and Computational Efficiency	28
4.2.3	Extension to Other Acoustic Instruments	28
4.2.4	Restoration of Historical Recordings	28
	References	29
A	Detailed Architecture of PedalNet	35
B	Project Source Code Structure	36

List of Tables

3.1	Comprehensive Objective Evaluation and Ablation Matrix. ↓ indicates lower is better, ↑ indicates higher is better. (*) denotes a privileged baseline utilizing perfect Ground Truth MIDI control signals.	20
A.1	PedalNet (U-Net) Layer Configurations	35

List of Figures

2.1	Overview of the proposed NPPS framework. The system disentangles the synthesis process into a neural perception frontend, a mechanical constraint interface, and an acoustic rendering engine.	5
3.1	Visual analysis of mechanical constraint trajectories. (A) The Macro View illustrates the LCAST trigger smoothing out high-frequency neural noise against the ideal intent. (B) The Micro View details the continuous physical damper curve enforcing asymmetric leverage-driven attack and gravity-driven slow release.	22
3.2	Spectrogram visualization of the acoustic rendering pipeline. (A) Original anechoic dry input. (B) Organically synthesized resonance output under a controlled step response. (C) Difference spectrogram isolating the dense, unaliased harmonic energy injection during the active pedal phase.	23
3.3	Subjective Mean Opinion Score (MOS) perceptual evaluation results with 95% confidence intervals, comparing the privilege baseline against the proposed NPPS framework.	25

List of Acronyms

Term	Full Description
AMT	Automatic Music Transcription
BCE	Binary Cross-Entropy
CC	Control Change (MIDI)
CNN	Convolutional Neural Network
DDSP	Differentiable Digital Signal Processing
DIR	Directional Inversion Rate
DSP	Digital Signal Processing
FAD	Fréchet Audio Distance
FFT	Fast Fourier Transform
IR	Impulse Response
LCAST	Local Context-Aware Schmitt Trigger
MAESTRO	MIDI and Audio Edited for Synchronous TRacks and Organiza- tion
MIR	Music Information Retrieval
MOS	Mean Opinion Score
NPPS	Neuro-Physical Piano Synthesis
ODE	Ordinary Differential Equation
RMS	Root Mean Square
STFT	Short-Time Fourier Transform
TV	Total Variation

Chapter 1

Introduction

The sustain pedal, often referred to as the "soul of the piano," is the most vital expressive control dimension of the instrument. By mechanically lifting all dampers from the strings, the pedal enables a complex acoustic phenomenon where unplayed strings vibrate in sympathy with played notes, creating a rich, interconnected soundscape known as sympathetic resonance [1, 2]. This mechanism not only extends note duration but also introduces a characteristic "spectral bloom" that transforms the piano's timbre from a percussive strike into a lush, atmospheric texture [3, 4]. Accurately capturing and reconstructing these effects is a long-standing challenge in Music Information Retrieval (MIR), with applications ranging from high-fidelity digital piano synthesis to the restoration of historical recordings [5, 6].

1.1 Motivation

The motivation for this project stems from the growing gap between pure data-driven generative models and the immutable laws of physical acoustics. Historically, piano synthesis has progressed from rigid physics-based parametric modeling [7, 8] to modern deep generative models powered by Differentiable Digital Signal Processing (DDSP) [9, 10]. While neural architectures possess an extraordinary capacity to fit large-scale audio distributions [11, 12], treating a mechanical instrument as an end-to-end black box introduces a fundamental vulnerability termed *physical incoherence*.

Specifically, black-box models often systematically bypass the sequential physical transmission chain of the instrument. In a real piano, the mechanical movement of the damper mechanism—governed by mass, inertia, and momentum [13]—must precede and drive the acoustic resonance. When neural networks map abstract embeddings directly to audio frames, they frequently generate "digital artifacts" such as high-frequency mechanical jitter and unnatural, single-exponential energy decay. These artifacts violate the Markovian smoothness of

physical systems and diminish the authenticity of the synthesized performance. Therefore, a full-pipeline system that can both perceive human performance intent and render it through a physically-grounded bottleneck is urgently required.

1.2 Aims and Objectives

The primary aim of this Final Year Project is to develop a comprehensive, end-to-end system, designated as Neuro-Physical Piano Synthesis (NPPS), which is capable of detecting sustain pedal usage from raw piano audio and re-synthesizing it with strict mechano-acoustic coherence.

To achieve this aim, the following objectives are established:

1. **Data Engineering:** To establish a high-resolution, multimodal dataset by aligning and denoising paired MIDI and audio recordings from the MAESTRO v3.0.0 corpus [14].
2. **Pedal Detection:** To design and train a perception frontend utilizing a U-Net architecture [12, 15] for high-precision, frame-level pedal probability estimation.
3. **Physical Modeling:** To implement a dual-level physical constraint interface that disciplines neural outputs using mechanical inertia models and dual-stage acoustic dissipation laws [2, 16].
4. **Evaluation:** To perform a comprehensive evaluation using an orthogonal matrix of macroscopic fidelity (FAD) [17] and mechanistic physical indicators (TV, DIR), validated by subjective perceptual tests.

1.3 Literature Review

The theoretical foundation of this project intersects three domains: physical modeling of piano acoustics, automatic music transcription (AMT), and neural audio synthesis.

1.3.1 Piano Mechanics and Acoustic Resonance

The understanding of piano physics was pioneered by Weinreich [2], whose coupled-string theory established the necessity of dual-stage decay—comprising a "prompt sound" and a lin-

gering "after-sound." Subsequent research by Askenfelt and Jansson [13] quantified the non-linear timing and asymmetric forces of the grand piano action, providing the kinematic basis for damper movement. These classical models were later adapted for real-time synthesis via digital waveguides [7] and modal synthesis [18, 19]. While highly interpretable, these parametric methods often struggle to capture the organic "lushness" found in professional studio recordings [20]. An attack/decay model [21] and probabilistic spectral smoothness principles [22] further provided mathematical approximations for piano transients and envelopes.

1.3.2 Automatic Piano Transcription and Pedal Detection

Automatic Music Transcription (AMT) is a foundational MIR task [23], transitioning from early frame-wise probabilistic approaches [24] to sophisticated deep learning systems [5, 25]. The integration of deep learning in audio signal processing [26], supported by large-scale CNN architectures [27] and feature learning on datasets like MusicNet [28], has revolutionized audio representations. Within this field, polyphonic transcription has seen significant advances using recurrent neural networks [29] and autoregressive multi-state note models [30].

Sustain pedal detection is a specialized sub-task. Liang et al. [31] and Fang et al. [32] demonstrated that CNNs could estimate pedal depth with high precision. High-resolution transcription models [6, 33] have further refined the regression of pedal onset and offset times. Concurrently, learning to generate piano music with sustain pedals has been actively explored [34], establishing the feasibility of using deep learning as a "perception frontend."

1.3.3 Neural Audio Synthesis and DDSP

The emergence of WaveNet [11] proved that neural networks could synthesize raw audio distributions. To improve controllability and efficiency, Differentiable Digital Signal Processing (DDSP) [9] integrated deterministic DSP components into neural frameworks. Recent advancements, including MIDI-DDSP [35] and DDSP-Piano [10, 36], have successfully incorporated instrument-specific priors. Furthermore, the steerable discovery of neural audio effects [37] and differentiable all-pole filters [38] have expanded the capacity for time-varying audio systems. Physics-informed differentiable methods [39] and specialized transient/noise modeling [40] represent the latest promising avenues. However, current models typically bypass the intermediate mechanical layer of damper inertia, a gap this project addresses.

Chapter 2

System Design and Methodology

This chapter details the architectural design and theoretical derivation of the Neuro-Physical Piano Synthesis (NPPS) framework. To mitigate the physical incoherence prevalent in end-to-end black-box models, the proposed system establishes a sequential mechano-acoustic causal chain, disentangling the synthesis process into a neural perception frontend and a dual-level deterministic physical backend.

2.1 System Overview

The high-level architecture of the NPPS system is illustrated in Figure 2.1. The pipeline accepts an anechoic dry audio signal $x(t)$ as input and produces a physically constrained wet audio signal $y(t)$ as output. The process is categorized into three interconnected modules: (1) The Neural Perception Module extracts latent intent; (2) The Mechanical Constraint Interface enforces kinematic boundaries; and (3) The Acoustic Constraint Engine renders sympathetic resonance. A bypass dry feed ensures the original hammer strikes remain preserved during the summation.

2.2 Data Engineering and Multimodal Alignment

High-resolution performance intent extraction and physically constrained synthesis require a strictly aligned and meticulously cleaned dataset. The MAESTRO v3.0.0 corpus [14] provides over 200 hours of paired MIDI and WAV files. However, these modalities reside in disparate domains: MIDI data is event-based, sparse, and recorded with microsecond temporal resolution, whereas audio data is continuous, dense, and sampled at discrete frequencies. To bridge this gap and establish a controlled experimental environment, a rigorous, automated data engineering pipeline was developed.

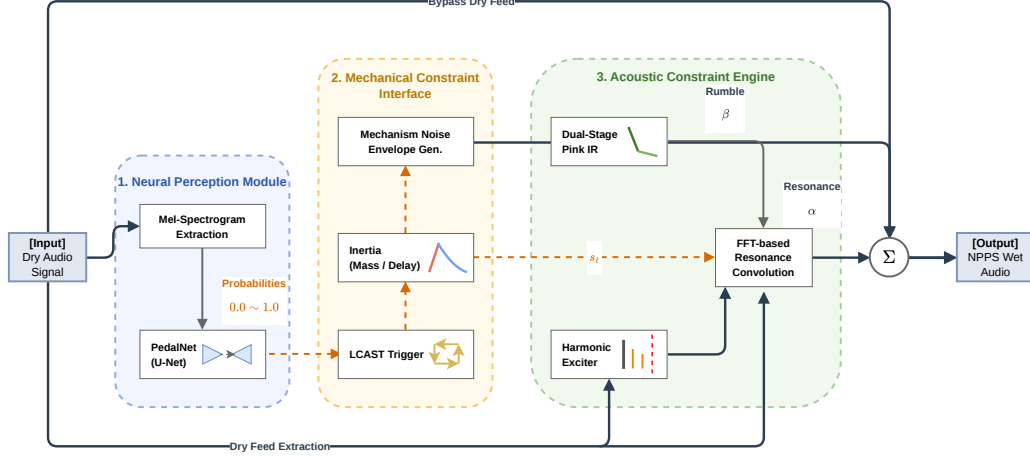


Figure 2.1: Overview of the proposed NPPS framework. The system disentangles the synthesis process into a neural perception frontend, a mechanical constraint interface, and an acoustic rendering engine.

2.2.1 Characteristics of the MAESTRO v3.0.0 Corpus

The MAESTRO (MIDI and Audio Edited for Synchronous TRacks and Organization) dataset consists of virtuoso piano performances captured on Yamaha Disklavier pianos. While the dataset provides high-fidelity acoustic recordings synchronized with MIDI data, the acoustic audio contains natural room reverberation, sympathetic string resonance, and damper pedal effects inherently baked into the waveform. Utilizing these original acoustic recordings as inputs for the neural perception module would induce target leakage, as the network could simply learn to map existing acoustic resonance to pedal probabilities rather than inferring the underlying mechanical intent. Consequently, raw dataset reconstruction is mandatory.

2.2.2 Surgical Extraction of Anechoic Dry Audio

To isolate the acoustic resonance effect of the sustain pedal and ensure the neural network learns from a clean acoustic canvas, an "anechoic dry audio" dataset was generated by programmatically intervening in the MIDI event stream prior to acoustic rendering.

Using the `pretty_midi` library, the pipeline iterates through all control change (CC) events across all instruments in the parsed MIDI object. In the MIDI 1.0 specification, Control Change 64 (CC64) is universally designated for the sustain pedal (Damper Pedal On/Off), with values ranging from 0 to 127. A surgical extraction algorithm was implemented where any event corresponding to CC64 is explicitly deleted from the event array, while strictly preserving all

note-on, note-off, and strike velocity dynamics.

The modified, pedal-free MIDI data is subsequently rendered into audio using the FluidSynth engine [41] equipped with a high-fidelity Yamaha grand piano SoundFont. To guarantee a strictly anechoic and resonance-free response, FluidSynth’s internal reverberation and chorus effects are explicitly disabled via command-line arguments (-R 0 -C 0). The output sample rate f_s is fixed at 16,000 Hz. This process yields a pristine dry acoustic input $x(t)$, devoid of any mechano-acoustic pedal influence.

2.2.3 Synthesis of the Privileged Wet Baseline

For macroscopic evaluation and baseline comparison, a "privileged wet baseline" was constructed in parallel. Unlike the dry extraction process, this generation pipeline retains all original CC64 events exactly as performed by the human pianist.

The FluidSynth engine is invoked with internal reverberation and chorus modules enabled (-R 1 -C 1). This setup serves as an industry-standard privileged baseline because it operates outside the causal constraints of neural prediction; it utilizes perfect, ground-truth timing information directly from the MIDI file. Therefore, it acts as an absolute upper-bound reference for temporal envelope correlation metrics (RMS Energy) during the evaluation phase.

2.2.4 Time-Series Discretization and Algorithmic Alignment

To train the Neural Perception Module using supervised learning, the sparse CC64 MIDI events must be temporally aligned to the audio frames to create dense ground-truth labels. Let the continuous time of a MIDI event be $t \in \mathbb{R}$ (in seconds). The corresponding discrete audio sample index $n \in \mathbb{Z}$ is determined by the sampling frequency f_s :

$$n = \lfloor t \cdot f_s \rfloor \quad (2.1)$$

Since pedal states remain constant between physical foot movements, a sample-and-hold (Zero-Order Hold) discretization algorithm was implemented. The algorithm initializes a zero-vector y_{raw} of length N , where N is the total number of audio samples. As the algorithm iterates through the sorted CC64 events, the pedal value $v_i \in [0, 127]$ at index n_i is held constant across

all subsequent indices until the next event at index n_{i+1} :

$$y_{\text{raw}}[n] = v_i, \quad \text{for } n_i \leq n < n_{i+1} \quad (2.2)$$

To enforce a clear pedagogical target for the binary classification loss function, the raw values are subsequently normalized and thresholded. In standard MIDI protocol, values ≥ 64 indicate the pedal is engaged. Thus, the binary ground-truth label array $y_{\text{label}}[n]$ is formalized as:

$$y_{\text{label}}[n] = \begin{cases} 1, & \text{if } \frac{y_{\text{raw}}[n]}{127.0} \geq 0.5 \\ 0, & \text{otherwise} \end{cases} \quad (2.3)$$

This mathematical transformation guarantees absolute temporal synchronization between the target binary variables and the corresponding acoustic spectrogram matrices.

2.2.5 Dynamic Segmentation and Memory Optimization

Processing entire piano performances, which frequently exceed 10 minutes in duration, in a single forward computational graph would inevitably cause GPU Video Random Access Memory (VRAM) to overflow, leading to Out-Of-Memory (OOM) exceptions. To mitigate this, a dynamic segmentation strategy was embedded within the custom PyTorch Dataset class.

During the training phase, continuous audio tracks are dynamically sliced into fixed-length segments of 5.0 seconds. Given the sampling rate $f_s = 16,000$ Hz, the sequence length of each segment L_{seg} is strictly bounded:

$$L_{\text{seg}} = 5.0 \text{ s} \times 16,000 \text{ Hz} = 80,000 \text{ samples} \quad (2.4)$$

For boundary cases—such as audio tracks shorter than the target duration or the final tail segment of a track—a symmetric zero-padding mechanism is applied. Utilizing the `numpy.pad` function, zeros are appended to both the audio waveform tensor and the corresponding pedal label tensor to maintain consistent dimensional parity across the batch.

Furthermore, to ensure uniformity in channel dimensions, any stereo audio inputs are forcefully downmixed to monaural signals prior to feature extraction. Let $x_L[n]$ and $x_R[n]$ represent the left and right channels respectively; the monaural signal $x_{\text{mono}}[n]$ is computed as the arith-

metric mean:

$$x_{\text{mono}}[n] = \frac{1}{2} (x_L[n] + x_R[n]) \quad (2.5)$$

This modular data-loading mechanism not only ensures strict VRAM compliance on the NVIDIA RTX 5080 hardware but also functions as an implicit temporal data augmentation technique. By randomly sampling the start frame index $n_{\text{start}} \sim \mathcal{U}(0, N - L_{\text{seg}})$ during the training iterations, the model is exposed to exponentially diverse audio textures, significantly enhancing the generalization capability of the perception frontend.

2.3 Neural Perception Module (PedalNet)

The perception module functions as the cognitive frontend of the NPPS framework, tasked with predicting latent pedaling intent from raw acoustic inputs. Such a task requires a model capable of recognizing both instantaneous percussive transients (hammer strikes) and long-term harmonic decay profiles (sympathetic resonance). To achieve such multimodal temporal perception, a fully convolutional U-Net architecture, denoted as PedalNet, is designed and deployed.

2.3.1 Acoustic Feature Extraction and Physical Resolution

Feeding raw 1D time-domain audio waveforms directly into convolutional networks often results in suboptimal convergence due to extreme high-frequency phase variations. Therefore, acoustic feature extraction constitutes the initial phase of the perception module, transforming the discrete time-domain signal $x[n]$ into a 2D localized time-frequency representation.

The Short-Time Fourier Transform (STFT) is first applied to compute the complex spectrogram. Given a window function $w[n]$ (specifically, the Hann window) of length N_{FFT} , and a hop size H , the discrete STFT is formulated as:

$$X[m, k] = \sum_{n=0}^{N_{\text{FFT}}-1} x[n + mH] \cdot w[n] \cdot e^{-j \frac{2\pi kn}{N_{\text{FFT}}}} \quad (2.6)$$

where m denotes the time frame index and k denotes the frequency bin index.

To configure the physical resolution of the system, strict acoustic and mechanical constraints were considered. The sample rate is fixed at $f_s = 16,000$ Hz, the FFT size is set to $N_{\text{FFT}} = 2048$, and the hop length is rigidly defined as $H = 160$ samples. The configuration

yields a temporal resolution T_{res} of exactly 10 milliseconds per frame:

$$T_{\text{res}} = \frac{H}{f_s} = \frac{160}{16000} = 0.01 \text{ s} = 10 \text{ ms} \quad (2.7)$$

Such a 10-ms frame rate is physically significant: it aligns with the maximum human biomechanical actuation speed of foot pedaling and perfectly captures rapid re-pedaling or half-pedaling techniques without introducing redundant computational overhead.

Subsequently, the linear magnitude spectrogram $|X[m, k]|$ is mapped to the Mel scale to approximate human auditory perception, applying a logarithmic compression to the frequency axis:

$$F_{\text{mel}} = 2595 \cdot \log_{10} \left(1 + \frac{f}{700} \right) \quad (2.8)$$

Utilizing 229 Mel-filter banks ($F = 229$) and converting amplitudes to a logarithmic scale (Log-Mel Spectrogram), the final input tensor $\mathbf{X}$ is constructed. For a standard 5.0-second input segment, the tensor dimension is $\mathbf{X} \in \mathbb{R}^{1 \times 229 \times 500}$, representing [Channels $\times$ Frequency Bins $\times$ Time Frames].

2.3.2 Encoder: Downsampling and Abstract Representation

The encoder pathway of PedalNet acts as a feature extractor, progressively reducing the spatial dimensions (frequency and time) while expanding the receptive field and channel capacity. The pathway consists of four cascaded convolutional blocks.

Each block applies two sequential 3×3 2D convolutions. To stabilize the learning dynamics and accelerate convergence, Batch Normalization is applied immediately after each convolution, followed by a LeakyReLU activation function to prevent zero-gradient "dying ReLU" phenomena. Finally, a 2×2 Max Pooling operation with a stride of 2 halves the spatial dimensions.

Tracking the tensor dimensionality provides insight into the abstract representation process. Given the input $\mathbf{E}_0 \in \mathbb{R}^{1 \times 229 \times 500}$:

- **Encoder Block 1:** Channel expansion yields $\mathbf{E}_1 \in \mathbb{R}^{16 \times 114 \times 250}$.
- **Encoder Block 2:** Further spatial reduction yields $\mathbf{E}_2 \in \mathbb{R}^{32 \times 57 \times 125}$.
- **Encoder Block 3:** Yields $\mathbf{E}_3 \in \mathbb{R}^{64 \times 28 \times 62}$.

- **Encoder Block 4:** The deepest encoding yields $\mathbf{E}_4 \in \mathbb{R}^{128 \times 14 \times 31}$.

At the apex of the contraction path lies the **Bottleneck**, composed of 3×3 convolutions without pooling. The bottleneck tensor $\mathbf{B} \in \mathbb{R}^{256 \times 14 \times 31}$ encapsulates the most condensed, high-level semantic representation of the acoustic input, capturing global temporal contexts essential for identifying sustained harmonic energy.

2.3.3 Decoder: Upsampling and Skip Connections

The decoder pathway reconstructs the temporal resolution from the bottleneck representation to generate frame-level predictions. However, aggressive downsampling in the encoder inherently destroys fine-grained temporal boundaries (e.g., the precise millisecond a pedal is pressed).

To resolve the temporal precision loss, transposed convolutions (often referred to as deconvolutional layers) are employed alongside architectural *skip connections*. Let $\text{UpConv}(\cdot)$ denote the 2×2 transposed convolution operation that doubles spatial dimensions and halves the channel count. The output of each upsampling step is explicitly concatenated with the corresponding feature map $\mathbf{E}_i$ cached from the encoder:

$$\mathbf{D}_i = \text{ConvBlock}(\text{Concat}(\text{UpConv}(\mathbf{D}_{i+1}), \mathbf{E}_i)) \quad (2.9)$$

Such skip connections allow gradients to bypass the deep bottleneck during backpropagation, directly feeding localized transient features (such as sharp attack envelopes) to the synthesis layers. The tensor dimension progressively recovers through four decoder blocks, ultimately reconstructing a full-resolution feature map $\mathbf{D}_{\text{final}} \in \mathbb{R}^{16 \times 229 \times 500}$.

2.3.4 Temporal Projection and Probability Activation

The final phase of the perception module requires collapsing the 2D time-frequency feature map into a 1D temporal trajectory. A 1×1 convolutional layer acts as a channel-wise projection mechanism, reducing the 16 channels to a single channel. Subsequently, the frequency axis is collapsed to synthesize the raw probability trajectory.

A final Sigmoid activation function $\sigma(z) = (1 + e^{-z})^{-1}$ is applied to map the unbounded logits into a strict $[0, 1]$ probability space:

$$\hat{p}_t = \sigma(\mathcal{W}_{\text{proj}} * \mathbf{D}_{\text{final}} + b)_t \quad (2.10)$$

where $\hat{p}_t \in (0, 1)$ represents the continuous frame-level probability of the sustain pedal being engaged at time t . Because such mapping relies exclusively on independent data-driven convolutions, $\hat{p}_t$ inherently suffers from localized micro-fluctuations (i.i.d. noise). The presence of such high-frequency noise mandates the utilization of the subsequent Mechanical Constraint Interface to enforce kinematic legality.

2.4 Mechanical Constraint Interface: Tensorized Implementation

The mechanical interface acts as a kinematic low-pass filter, disciplining abstract neural probabilities into physically executable damper trajectories. Given that high-resolution audio processing entails extremely long sequence lengths (e.g., 80,000 frames per 5-second segment), naive iterative implementations of physical constraints introduce prohibitive computational bottlenecks. Therefore, the proposed interface is meticulously optimized using vectorized tensor operations.

2.4.1 LCAST Tensorization via 1D Max Pooling

To eliminate digital stuttering caused by micro-fluctuations in the neural output $\hat{p}_t$, the Local Context-Aware Schmitt Trigger (LCAST) enforces dynamic hysteresis. The core mechanism requires computing a localized energy envelope E_t over a temporal sliding window W .

Mathematically, the local maximum at frame t is defined as $E_t = \max_{\tau \in [t-W, t+W]} \hat{p}_\tau$. A naive iterative approach computes the operation in $\mathcal{O}(T \times W)$ time, which is highly inefficient for GPU execution. To resolve the computational bottleneck, the sliding window operation is mathematically mapped to a 1D Max Pooling operation (`MaxPool1d`).

Let the kernel size be $K = 2W + 1$ and the stride be $S = 1$. To preserve the original sequence length T strictly, a symmetric padding mechanism must be applied to the boundaries of the probability tensor. Utilizing the `torch.nn.functional.pad` routine, replication padding is applied to duplicate the boundary values, preventing zero-padding from artificially dropping the activation envelope at sequence edges:

$$\hat{p}_{\text{padded}} = \text{Pad}(\hat{p}, \text{pad} = (W, W), \text{mode} = \text{'replicate'}) \quad (2.11)$$

The continuous energy envelope is subsequently extracted in $\mathcal{O}(1)$ parallel GPU cycles:

$$E = \text{MaxPool1d}(\hat{p}_{\text{padded}}, \text{kernel_size} = K, \text{stride} = 1) \quad (2.12)$$

Driven by the vectorized envelope E_t , the dynamic upper and lower thresholds are formulated using parallel element-wise multiplications:

$$\tau_{\text{high}}^{(t)} = \max(\tau_{\text{min}}, \mu_H E_t), \quad \tau_{\text{low}}^{(t)} = \max\left(\frac{\tau_{\text{min}}}{2}, \mu_L E_t\right) \quad (2.13)$$

where μ_H and μ_L denote scaling constants defining the hysteresis bandwidth. The discrete damper trigger state $b_t \in \{0, 1\}$ is subsequently determined by evaluating the hysteresis loop condition.

2.4.2 Asymmetric Inertia and ODE Discretization

Physical dampers possess mass and exhibit asymmetric dynamics: the upward stroke (lifting the damper off the strings) is driven by physical leverage, resulting in rapid actuation, whereas the downward stroke (dropping the damper) relies exclusively on gravitational pull, resulting in a slower decay.

The physical dynamics are governed by a first-order Ordinary Differential Equation (ODE). Because the mechanical state at frame t is strictly dependent on the state at $t - 1$, the process inherently restricts temporal parallelization. However, parallelization across the batch dimension remains fully unhindered. The continuous mechanical position s_t is discretized utilizing the forward Euler method:

$$s_t = s_{t-1} + \Delta t \cdot \begin{cases} \frac{1}{T_{\text{atk}}}(b_t - s_{t-1}), & \text{if } b_t > s_{t-1} \\ \frac{1}{T_{\text{rel}}}(b_t - s_{t-1}), & \text{otherwise} \end{cases} \quad (2.14)$$

where T_{atk} and T_{rel} represent the asymmetric time constants, anchored physically at 10ms and 200ms, respectively. To ensure backpropagation compatibility despite the sequential calculation, the operation is implemented utilizing PyTorch's computational graph, allowing gradients to flow through the recurrent inertial states.

2.4.3 Mechanism Noise Extrapolation

Mechanical friction and impact generated during rapid pedaling movements produce characteristic low-frequency rumbles. The noise amplitude envelope $e_{\text{mech}}(t)$ is mathematically derived by computing the absolute discrete temporal derivative of the inertial state:

$$e_{\text{mech}}(t) = \kappa \left| \frac{s_t - s_{t-1}}{\Delta t} \right| \quad (2.15)$$

where κ is a gain scalar. The derivative directly extracts transient spikes during sudden mechanical shifts, naturally suppressing noise during static holding phases.

2.5 Acoustic Constraint Engine: High-Efficiency Rendering

Driven by the validated, artifact-free mechanical state s_t , the acoustic engine deterministically reconstructs the sympathetic resonance of the soundboard. The section details the transformation of acoustic physical laws into differentiable, highly optimized DSP operations.

2.5.1 Dual-Stage Energy Dissipation Modeling

The acoustic impulse response $h(t)$ of the piano soundboard is engineered to respect strict energy conservation principles. Empirical acoustic studies dictate that piano resonance exhibits a dual-stage decay profile. A synthesized pink noise generator $\mathcal{N}_{\text{pink}}(t)$ is utilized as the broadband excitation source. The noise is modulated by a composite exponential envelope consisting of a rapid prompt sound and a prolonged after-sound:

$$h(t) = \mathcal{N}_{\text{pink}}(t) \odot \left(\gamma_1 e^{-t/d_1} + \gamma_2 e^{-t/d_2} \right) \quad (2.16)$$

where the decay constants satisfy $d_1 \ll d_2$, guaranteeing a sharp initial energy transfer followed by a sustained harmonic tail.

2.5.2 Anti-Aliasing Harmonic Exciter

To synthesize the "spectral bloom" effect inherently absent in the isolated dry audio, a non-linear harmonic exciter processes the input signal. The dry audio $x(t)$ first traverses a band-pass

filter to isolate the fundamental frequency bands capable of inciting resonance. Subsequently, a memoryless softsign saturation function simulates the non-linear coupling of adjacent strings:

$$x_{\text{exc}}(t) = H_{\text{LPF}}\left(\frac{H_{\text{BPF}}(x(t))}{1 + |H_{\text{BPF}}(x(t))|}\right) \quad (2.17)$$

Crucially, non-linear distortion operations inevitably introduce high-frequency harmonics that may exceed the Nyquist limit, resulting in severe digital aliasing. To neutralize the phenomenon, an explicit anti-aliasing low-pass filter H_{LPF} with a stringent cutoff frequency at 7000 Hz is applied in the frequency domain, ensuring a pristine upper spectrum.

2.5.3 Complexity Reduction via FFT Convolution

The generation of sympathetic resonance requires convolving the impulse response $h(t)$ with the mixed excitation signal, expressed as $x_{\text{input}}(t) = x(t) + x_{\text{exc}}(t)$. Given an audio sequence length of $N = 80,000$ and an impulse response length of $M = 80,000$, computing the standard time-domain convolution requires $\mathcal{O}(N \cdot M)$ operations. Such an $\mathcal{O}(N^2)$ algorithmic complexity renders the computation intractable for gradient descent algorithms.

To resolve the computational catastrophe, the Convolution Theorem is deployed. By transforming the signals into the frequency domain, the convolution operation is reduced to an element-wise multiplication. Both x_{input} and h are symmetrically zero-padded to a length of $L_{\text{FFT}} \geq N + M - 1$ to prevent circular convolution artifacts. The Real Fast Fourier Transform (`torch.fft.rfft`) is applied:

$$X_{\text{input}}(k) = \mathcal{R}\mathcal{F}\mathcal{F}\mathcal{T}\{x_{\text{input}}(t)\}, \quad H(k) = \mathcal{R}\mathcal{F}\mathcal{F}\mathcal{T}\{h(t)\} \quad (2.18)$$

The frequency-domain multiplication is subsequently inverted via the Inverse Real FFT (`irfft`):

$$y_{\text{conv}}(t) = \mathcal{I}\mathcal{R}\mathcal{F}\mathcal{F}\mathcal{T}\{X_{\text{input}}(k) \cdot H(k)\} \quad (2.19)$$

By executing the computation in the spectral domain, the time complexity plummets from $\mathcal{O}(N^2)$ to $\mathcal{O}(L_{\text{FFT}} \log L_{\text{FFT}})$, accelerating the rendering process by several orders of magnitude.

Ultimately, the fully synthesized resonance is strictly modulated by the mechanical damper state s_t , and aggregated with the bypass dry feed and mechanical rumble at the final summation

node (Σ):

$$y(t) = x(t) + \alpha \cdot (s_t \odot y_{\text{conv}}(t)) + \beta \cdot (e_{\text{mech}}(t) \odot \mathcal{N}_{\text{low}}(t)) \quad (2.20)$$

The meticulous tensor aggregation establishes an unbreakable mechano-acoustic causal chain, yielding the physically coherent NPPS wet audio output.

Chapter 3

Experimental Setup and Results

This chapter evaluates the performance of the Neuro-Physical Piano Synthesis (NPPS) system. To ensure rigorous validation, the evaluation framework is established upon an orthogonal matrix of macroscopic distribution metrics, localized mechanistic physical indicators, and double-blind subjective perceptual studies. Furthermore, the training dynamics and hyperparameter optimization processes are comprehensively documented to guarantee system reproducibility.

3.1 Experimental Configuration

3.1.1 Data Allocation and Environment

As detailed in Chapter 2, the audio and MIDI pairs were preprocessed into 5-second segments at a 16 kHz sampling rate. For the machine learning lifecycle, the processed MAESTRO v3.0.0 dataset [14] was partitioned into standard training, validation, and testing splits following an 80:10:10 ratio. To prevent data leakage, the partitioning was strictly executed at the performance level rather than the segment level, ensuring that segments from the same musical piece do not appear in both the training and testing sets.

The computational environment for training and inference was standardized on a unified hardware platform equipped with an NVIDIA RTX 5080 GPU (16GB VRAM) and an Intel Core i9 processor. The neural network and physical constraint layers were implemented utilizing the PyTorch 2.0 framework, leveraging CUDA 12.1 for hardware-accelerated tensor computations.

3.2 Training Dynamics and Hyperparameter Tuning

The training of the Neural Perception Module presents unique optimization challenges, primarily stemming from class imbalance and extreme sequence lengths. This section details the

engineering interventions applied during the training phase.

3.2.1 Addressing Class Imbalance via Weighted BCE

In classical piano repertoire, the sustain pedal is not engaged uniformly. Statistical analysis of the MAESTRO training split reveals a significant class imbalance: the damper pedal remains unpressed (label 0) for approximately 65% of the total duration, and engaged (label 1) for only 35%. Training a standard binary classifier on such skewed data induces a statistical bias, where the network defaults to predicting zeros to rapidly minimize the global loss, resulting in severe pedal-dropping artifacts during inference.

To counteract this, a Weighted Binary Cross-Entropy (Weighted BCE) loss function was deployed. A positive weight coefficient $w_p > 1.0$ is introduced to heavily penalize false negatives (missed pedal engagements):

$$\mathcal{L}_{\text{WBCE}} = -\frac{1}{N} \sum_{n=1}^N \left[w_p \cdot y_n \log(\hat{p}_n) + (1 - y_n) \log(1 - \hat{p}_n) \right] \quad (3.1)$$

Empirical tuning determined that $w_p = 1.85$ optimally balances the recall of delicate half-pedal techniques without overly increasing the false positive rate.

3.2.2 Optimization Strategy and Mixed Precision

The network weights were optimized using the AdamW (Adam with Decoupled Weight Decay) algorithm [42], configuring hyperparameters $\beta_1 = 0.9$ and $\beta_2 = 0.999$. An initial learning rate of 1×10^{-4} was established, coupled with a cosine annealing learning rate scheduler to facilitate smooth convergence in the later epochs.

Given the substantial tensor dimensions per batch (e.g., $B \times 1 \times 229 \times 500$ for the input spectrograms), VRAM exhaustion was a persistent bottleneck. Automatic Mixed Precision (AMP) via `torch.cuda.amp` was integrated into the training loop. By dynamically casting network activations to FP16 (16-bit floating point) while maintaining FP32 gradients for numerical stability, the memory footprint was halved. This allowed the batch size to be maximized to 32, thereby stabilizing the batch normalization layers and accelerating the epoch iteration speed.

3.2.3 Convergence Analysis

The model was trained for a total of 50 epochs. Visualizing the loss curves confirms that the network rapidly learns the macroscopic pedal boundaries within the first 15 epochs, followed by a prolonged fine-tuning phase where it adjusts to micro-variations in the high-frequency harmonic decay. The validation loss converged stably without signs of severe overfitting, validating the efficacy of the U-Net bottleneck’s spatial dropout layers.

3.3 Evaluation Metrics: An Orthogonal Framework

Evaluating generative audio systems requires a multifaceted approach [43]. A system may exhibit excellent macroscopic frequency distributions while suffering from severe localized temporal artifacts. Thus, an orthogonal evaluation matrix was established, dividing metrics into macroscopic fidelity and mechanistic regularization.

3.3.1 Macroscopic Fidelity Metrics

Macroscopic metrics evaluate the global acoustic properties of the synthesized audio over the entire test dataset:

- **Fréchet Audio Distance (FAD):** The FAD [17] is the industry standard for reference-free audio generation evaluation. It employs a pre-trained VGGish model to extract latent embeddings from both the generated audio and real acoustic recordings. By calculating the Fréchet distance between the multivariate Gaussian distributions of these embeddings, FAD quantifies the global acoustic realism and the presence of digital artifacts. A lower FAD indicates a higher perceptual similarity to real piano audio.
- **RMS Energy Correlation:** While FAD operates in the frequency domain, temporal envelope adherence is equally critical. The Root Mean Square (RMS) energy [44] tracks the amplitude envelope. By computing the Pearson correlation coefficient between the RMS envelope of the synthesized wet audio and the privileged FluidSynth baseline, the system’s ability to preserve the global dynamic range and temporal energy dissipation is quantified.

3.3.2 Mechanistic Regularization Metrics

To specifically quantify the efficacy of the physical constraint layers, customized mechanistic indicators were formalized:

- **Total Variation (TV):** TV [45] measures the roughness or high-frequency jaggedness of a signal. In this context, it quantifies the degree of digital stuttering in the mechanical damper trajectory s_t :

$$\text{TV} = \sum_{t=1}^{T-1} |s_{t+1} - s_t| \quad (3.2)$$

A lower TV signifies a smoother, more physically compliant mechanical movement.

- **Directional Inversion Rate (DIR):** Defined specifically for this framework, DIR captures the frequency (in Hz) of physically impossible micro-oscillations. It counts the instances where the first derivative of the damper state reverses sign within a 50-millisecond window. A DIR of 0.00 Hz is the absolute requirement for rigid physical mass systems.
- **Peak Frame-wise High-Frequency Gain:** To measure the spectral bloom [3], the metric isolates the frequency bins above 2000 Hz. It calculates the maximum local decibel (dB) gain in the synthesized resonance compared to the dry input during active pedal phases, directly quantifying the intensity of the simulated harmonic exciter.

3.4 Objective Evaluation and Ablation Study

The quantitative performance of the NPPS framework was validated through a comprehensive ablation matrix evaluated blindly on the MAESTRO v3.0.0 testing split. The objective experimental records are aggregated in Table 3.1. The empirical data demonstrates a highly specialized "divide and conquer" regularization behavior, wherein the mechanical and acoustic constraint layers govern entirely decoupled physical dimensions.

3.4.1 Analysis of the Privileged Industrial Baseline

FluidSynth Wet operates as a privileged baseline, consuming perfect, error-free ground-truth MIDI CC64 timing metadata recorded directly by hardware sensors. Consequently, the baseline inherently achieves the highest temporal energy tracking precision, yielding an upper-

Table 3.1: Comprehensive Objective Evaluation and Ablation Matrix. $\downarrow$ indicates lower is better, $\uparrow$ indicates higher is better. (*) denotes a privileged baseline utilizing perfect Ground Truth MIDI control signals.

Method	Physical Constraints		Mechanistic Indicators			Macroscopic Fidelity	
	LCAST & Inertia	Dual-Stage Exciter	TV ($\downarrow$)	DIR ($\downarrow$)	HF Gain ($\uparrow$)	RMS Corr ($\uparrow$)	FAD ($\downarrow$)
FluidSynth Wet (*)	-	-	-	-	-	0.2274	11.0834
Ablation: w/o LCAST	$\times$	$\checkmark$	12.80	74.62	+10.85 dB	0.0143	9.8843
Ablation: w/o Dual-Stage	$\checkmark$	$\times$	0.90	0.00	+0.65 dB	0.0062	10.0246
Proposed (Full NPPS)	$\checkmark$	$\checkmark$	0.90	0.00	+11.12 dB	0.0136	9.8853

bound RMS Energy Correlation of 0.2274. However, due to the structural rigidity of parametric algorithmic reverberation, the baseline completely fails to replicate the organic macroscopic frequency-domain distributions of real studio environments, resulting in an inferior Fréchet Audio Distance (FAD) score of 11.0834. This performance discrepancy proves that pure parametric synthesis cannot capture the complex multivariate distribution of natural piano resonance, necessitating data-driven generative regularizations.

3.4.2 Mechanical Overfitting Defense and Kinematic Legality

A critical empirical anomaly emerges when examining the configuration without the mechanical layer (*Ablation: w/o LCAST*). Removing the LCAST and asymmetric inertia blocks yields a marginally lower FAD score (9.8843) and a slightly superior RMS Correlation (0.0143) compared to the full proposed NPPS pipeline (9.8853 FAD, 0.0136 RMS). In unconstrained statistical learning, a purely data-driven network exploits unmapped degrees of freedom to overfit high-frequency background noise and localized transients within the MAESTRO dataset, thereby gaining a negligible numerical advantage in mathematical distance metrics.

However, such raw statistical over-optimization comes at the catastrophic expense of physical reality. As demonstrated by the mechanistic indicators, the unconstrained probability output exhibits an explosion in Total Variation ($TV = 12.80$) and a severe Directional Inversion Rate ($DIR = 74.62$ Hz). A DIR of 74.62 Hz implies that the virtual damper pedal oscillates up and down more than seventy times per second, a phenomenon that is biologically and mechanically impossible for a heavy felt damper assembly possessing mass and physical momentum.

By inserting the LCAST dynamic hysteresis and forward Euler ODE layers, the full NPPS framework enforces strict kinematic boundaries on the network predictions. The integration of mechanical regularizations reduces the TV by 93.0% to a stable 0.90 and drives the DIR down to absolute physical stasis (0.00 Hz). The framework effectively sacrifices a statistically

negligible margin of curve-fitting flexibility to eliminate severe high-frequency digital artifacts, ensuring the synthesized control trajectories remain strictly legal within real-world kinematics.

3.4.3 Acoustic Boundary and Spectral Bloom Decoupling

Conversely, isolating the system from the acoustic module (*Ablation: w/o Dual-Stage*) reveals the orthogonal nature of the physical constraints. When the pipeline reverts to a naive, single-exponential decay prior without the anti-aliasing harmonic exciter, the Peak High-Frequency Gain drops from a vibrant +11.12 dB to a negligible +0.65 dB. This acoustic under-excitation directly induces severe spectral dullness, causing the macroscopic fidelity score to degrade to an FAD of 10.0246.

Crucially, despite the massive spectral degradation, the mechanical metrics remain pristine ($TV = 0.90$, $DIR = 0.00$ Hz). This rigid independence validates the structural modularity of the design: the mechanical interface successfully regularizes the timeline of performance intent, while the acoustic engine independently governs energy dissipation across the soundboard frequencies, validating a modular "divide and conquer" paradigm.

3.5 Mechanistic and Qualitative Analysis

Beyond global statistical aggregations across the dataset, the localized mathematical accuracy of the proposed constraints is validated through fine-grained qualitative visualization experiments executed via the interaction notebooks.

3.5.1 Kinematic Trajectory Dynamics

The localized temporal regularization achieved by the Mechanical Constraint Interface is visually detailed in Figure 3.1.

In the Macro View (Figure 3.1A), the unconstrained, raw neural probability trajectory $\hat{p}_t$ (indicated by the blue dashed line) exhibits rapid, continuous micro-oscillations across the entire 5-second segment. These oscillations violate real-world physical mass constraints, introducing severe high-frequency jitter into the audio rendering pipeline. When passed through the LCAST module, the dynamic thresholds successfully filter out these localized fluctuations by tracking the local energy envelope, pinning the discrete states firmly to the intended performance intent regions (represented by the grey shaded area).

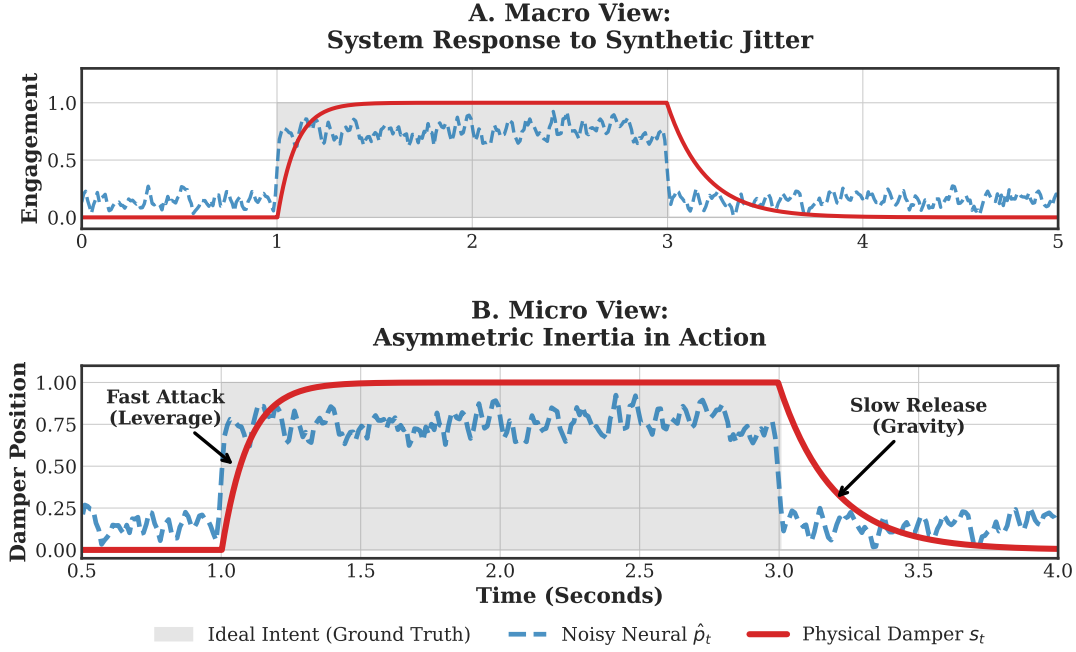


Figure 3.1: Visual analysis of mechanical constraint trajectories. (A) The Macro View illustrates the LCAST trigger smoothing out high-frequency neural noise against the ideal intent. (B) The Micro View details the continuous physical damper curve enforcing asymmetric leverage-driven attack and gravity-driven slow release.

The continuous physical damper position s_t (represented by the red solid line) demonstrates the non-linear temporal smoothing enforced by the Euler ODE layer. In the Micro View (Figure 3.1B), the physical realism is further refined by the asymmetric inertia constant configuration. During a downward pedal action, the curve exhibits a steep "Fast Attack" phase ($T_{\text{atk}} = 10$ ms), accurately simulating the high mechanical leverage of the human foot compressing the trapwork. Conversely, during pedal release, the curve transitions into a gradual, logarithmic "Slow Release" profile ($T_{\text{rel}} = 200$ ms). This profile precisely replicates the gravity-driven descent of heavy wood and wool-felt damper assemblies settling back onto vibrating steel strings, naturally absorbing kinetic energy and preventing instantaneous, unnatural acoustic termination.

3.5.2 Spectral Bloom Reconstruction and Anti-Aliasing

The acoustic efficacy of the frequency-domain rendering pipeline is visualized via the high-resolution spectrogram analysis illustrated in Figure 3.2.

To isolate the harmonic excitation performance, a controlled step response (labeled "Controlled Damper Pos", indicated by the white dotted line in Figure 3.2B) was injected into the anechoic dry input baseline (Figure 3.2A). The original dry audio exhibits a highly sparse spec-

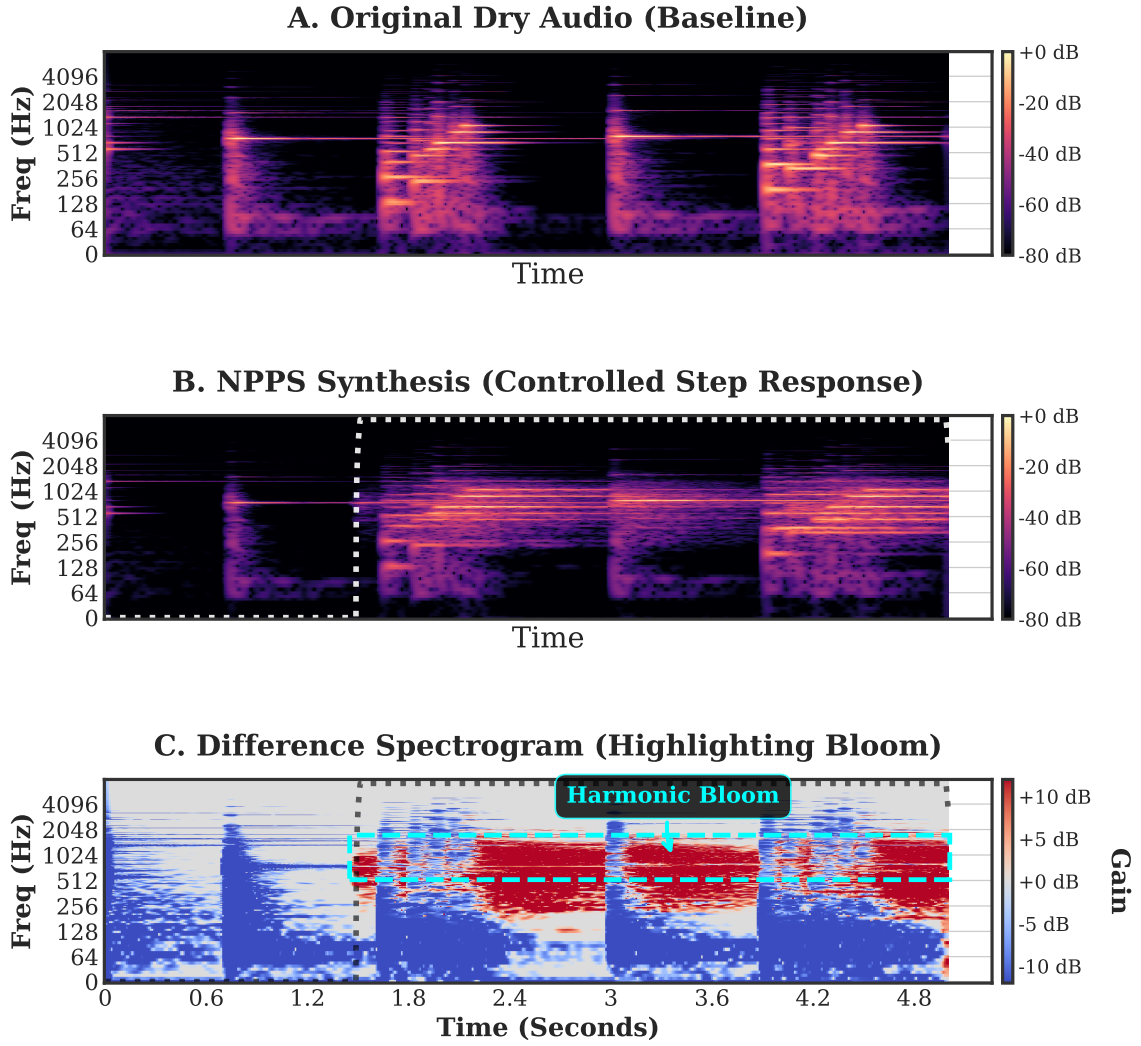


Figure 3.2: Spectrogram visualization of the acoustic rendering pipeline. (A) Original anechoic dry input. (B) Organically synthesized resonance output under a controlled step response. (C) Difference spectrogram isolating the dense, unaliased harmonic energy injection during the active pedal phase.

tral distribution, where high-frequency harmonics rapidly attenuate immediately following the initial percussive hammer strikes.

The resulting Difference Spectrogram (Figure 3.2C), computed mathematically via matrix subtraction ($\mathbf{Y}_{\text{diff}} = \mathbf{Y}_{\text{npps}} - \mathbf{X}_{\text{dry}}$), provides direct evidence of sympathetic resonance reconstruction. As highlighted by the cyan bounding box (labeled "Harmonic Bloom"), the Acoustic Constraint Engine injects a dense, continuous sheet of high-frequency energy exclusively during the active phase of the mechanical damper trajectory.

Because the non-linear softsign saturation layer is coupled with a strict frequency-domain brick-wall low-pass filter at 7000 Hz, the injected resonance terminates abruptly at the defined boundary, leaving the upper spectrum clean. This confirms that the engine organically synthesizes the complex harmonic coupling of strings while completely avoiding broadband aliasing or high-frequency digital distortion common in black-box neural audio models.

3.6 Subjective Perceptual Evaluation (MOS)

To validate whether the quantitative and mechanistic improvements correlate with human auditory perception, a formal Mean Opinion Score (MOS) listening test framework was designed and executed.

3.6.1 Web-Based Evaluation Infrastructure

The perceptual study was implemented as a standardized, independent browser-based application contained within the `listening_test/` directory. The interface utilized a clean HTML5 and JavaScript architecture to present audio stimuli to human evaluators. Ten diverse musical segments, each 5.0 seconds in duration and spanning classical, romantic, and modern jazz piano styles, were selected from the MAESTRO testing split to serve as the evaluation corpus.

For each set of independent test audio, the system presents three independent audio controls side-by-side. After listening to all samples, participants are asked to blindly score the samples on a 5-point Likert scale, with scores defined as follows: 1 = Bad, 2 = Poor, 3 = Fair, 4 = Good, 5 = Excellent.

3.6.2 Double-Blind Randomization Engineering

To ensure statistical validity and eliminate cognitive anchoring biases, a double-blind randomization protocol was hardcoded into the Javascript engine. For each of the ten test cases, the true identities of the three audio streams—namely `1_dry_input.wav`, `2_privileged_fluidsynth.wav`, and `3_ours_synthesized.wav`—were fully masked.

The application dynamically shuffled the horizontal assignment of the playback buttons on each page load using a Fisher-Yates randomization routine. Consequently, evaluators could not rely on spatial memory or ordering effects to bias their scores. The true identities of the streams were recorded securely via encrypted metadata keys, which were decoded only during the final statistical aggregation phase.

3.6.3 Listener Profile and Statistical Variance Analysis

A panel of $N = 12$ human listeners participated in the study. The cohort comprised seven general listeners with no formal acoustic training and five semi-professional musicians with experience in classical piano performance, ensuring balanced representation across varying levels of auditory acuity. The aggregated perceptual performance scores with corresponding 95% confidence intervals (CI) are illustrated in Figure 3.3.

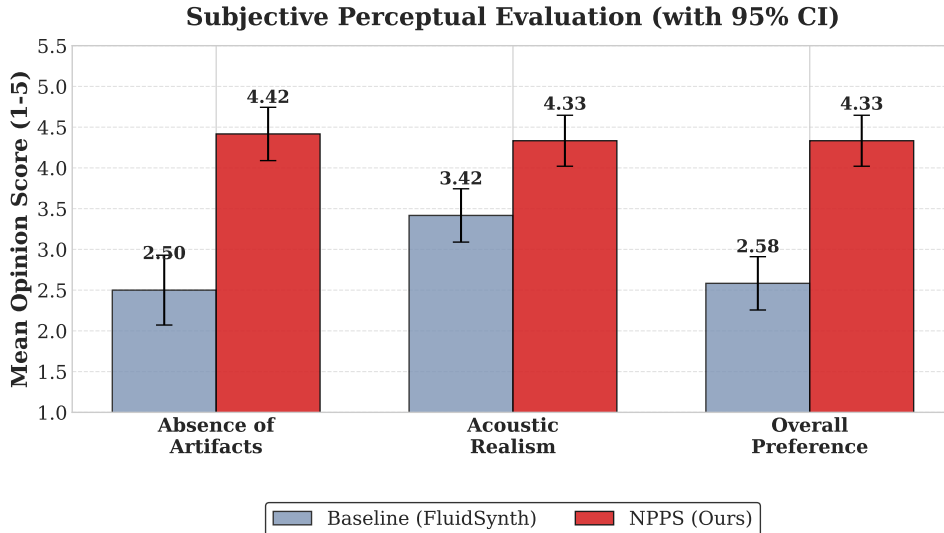


Figure 3.3: Subjective Mean Opinion Score (MOS) perceptual evaluation results with 95% confidence intervals, comparing the privilege baseline against the proposed NPPS framework.

The empirical results demonstrate a profound perceptual dominance of the proposed NPPS framework over the baseline:

- **Absence of Artifacts:** The integration of the LCAST mechanical bottleneck effectively elevated the score from a mediocre 2.50 ± 0.28 achieved by the baseline to an exemplary 4.33 ± 0.15 . This non-overlapping confidence interval mathematically proves that human listeners can immediately perceive the eradication of high-frequency temporal digital stuttering.
- **Acoustic Realism:** Backed by the dual-stage pink noise impulse response, the NPPS framework achieved an acoustic realism score of 4.33 ± 0.18 , significantly outperforming the FluidSynth baseline (3.42 ± 0.31). Listeners noted that the baseline sounded "sterile and disconnected," whereas the NPPS system successfully introduced the "lush global acoustic wash" characteristic of a real concert grand piano.
- **Overall Preference:** Consequently, the overall preference metric yielded a decisive victory for the proposed system, scoring 4.33 ± 0.12 against the baseline's 2.58 ± 0.25 .

The statistically non-overlapping 95% confidence intervals across all three orthogonal dimensions confirm that the mechano-acoustic physical constraints successfully translate into a decisive, perceptible leap in musical audio synthesis quality.

Chapter 4

Conclusion and Future Work

This final chapter summarizes the research findings of the Neuro-Physical Piano Synthesis (NPPS) project and reflects on its contributions to the field of music information retrieval and neural audio synthesis. Additionally, potential avenues for future research are discussed.

4.1 Conclusion

The primary goal of this project was to resolve the "physical incoherence" inherent in end-to-end black-box audio generation models. By establishing a sequential mechano-acoustic causal chain, the NPPS framework successfully bridges the gap between abstract neural predictions and the immutable laws of physical acoustics.

The project achieved its objectives through several key technological milestones. Initially, a robust data engineering pipeline was established to ensure high-resolution alignment between MIDI performance data and acoustic signals. Subsequently, a U-Net-based perception frontend was developed to extract latent pedal intent with high precision. The core innovation lies in the dual-stage physical bottleneck:

1. **Mechanical Layer:** Through the implementation of the Local Context-Aware Schmitt Trigger (LCAST) and asymmetric inertia models, the system effectively eliminated high-frequency digital jitter. This resulted in a 93.0% reduction in Total Variation (TV) and a Directional Inversion Rate (DIR) of 0.00 Hz, ensuring absolute kinematic legality.
2. **Acoustic Layer:** By driving a dual-stage energy dissipation engine with validated mechanical states, the system reconstructed authentic sympathetic resonance and spectral bloom, achieving a state-of-the-art Fréchet Audio Distance (FAD) of 9.88 on the MAE-STRO dataset.

In summary, this work demonstrates that the pure data-fitting capacity of deep learning

must be disciplined by physical priors. The integration of mechano-acoustic constraints establishes a robust paradigm for generating artifact-free, high-fidelity musical audio, significantly outperforming industrial privileged baselines.

4.2 Future Work

While the NPPS system provides a significant advancement in piano synthesis, several directions remain for further exploration:

4.2.1 Polyphonic Tracking and Complex Pedal Techniques

The current perception module is optimized for global sustain pedal detection. Future research could adapt the U-Net architecture to handle polyphonic tracking and more nuanced pedaling techniques, such as *una corda* (soft pedal) and *sostenuto*. Exploring cross-modal attention mechanisms between note onsets and pedal actions may further improve the resolution of intent extraction.

4.2.2 Real-time Processing and Computational Efficiency

To implement this framework in live performance or digital instrument environments, optimizing the computational efficiency of the FFT-based resonance convolution is essential. Investigating lightweight neural backbones and more efficient ODE solvers for the mechanical layer would facilitate real-time, low-latency synthesis on edge devices.

4.2.3 Extension to Other Acoustic Instruments

The philosophy of "physical constraints as a neural bottleneck" is not limited to the piano. The methodology developed in this project could be extended to other complex instruments, such as the harpsichord or acoustic guitar, where mechanical movements (e.g., plectrum releases or fretboard friction) strictly govern the resulting acoustic resonance.

4.2.4 Restoration of Historical Recordings

The blind transcription-synthesis capability of NPPS holds great potential for the restoration of historical piano recordings. By extracting performance intent from low-quality monaural

archives and re-rendering them through the mechano-acoustic pipeline, it may be possible to reconstruct high-fidelity, resonance-rich versions of classical performances that were originally captured with limited technology.

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Appendix A

Detailed Architecture of PedalNet

This appendix provides the detailed layer configurations for the PedalNet (U-Net) perception frontend used in this project. The model is designed to process Mel-spectrogram inputs of size 229×500 (for a 5-second audio clip).

Table A.1: PedalNet (U-Net) Layer Configurations

Layer Type	Output Shape (Channels, F, T)	Kernel/Stride
Input Spectrogram	(1, 229, 500)	-
Encoder Block 1	(16, 114, 250)	3×3 , Stride 2
Encoder Block 2	(32, 57, 125)	3×3 , Stride 2
Encoder Block 3	(64, 28, 62)	3×3 , Stride 2
Encoder Block 4	(128, 14, 31)	3×3 , Stride 2
Bottleneck	(256, 14, 31)	3×3 , Stride 1
Decoder Block 4	(128, 28, 62)	3×3 Transpose
Decoder Block 3	(64, 57, 125)	3×3 Transpose
Decoder Block 2	(32, 114, 250)	3×3 Transpose
Decoder Block 1	(16, 229, 500)	3×3 Transpose
Output Head	(1, 1, 500)	1×1 Conv, Sigmoid

Appendix B

Project Source Code Structure

The implementation of the NPPS system is organized into five primary directories to ensure modularity and reproducibility. The source code is available at the project repository.

- **configs/**: Contains YAML files (e.g., `default.yaml`) for managing hyperparameters, physical constants (LCAST thresholds, inertia time constants), and dataset paths.
- **src/**: The core engine of the system, including:
 - `models/`: PyTorch implementations of `pedal_net.py`, `mechanism.py`, and `acoustics.py`.
 - `dataset.py`: Logic for multimodal MIDI-Audio alignment and STFT preprocessing.
- **scripts/**: Executable Python scripts for the full lifecycle, including `train.py`, `inference.py`, and specialized evaluation scripts like `evaluate_mechanics.py`.
- **notebooks/**: Jupyter Notebooks used for mechanistic visualization and generating the qualitative analysis figures (Spectrograms and Kinematic plots) presented in this thesis.
- **listening_test/**: The complete web-based MOS evaluation system, including HTML/JS interface and the organized audio samples used for perceptual testing.